ATMEGA 4809 µC

U7
ATMEGA4809-AUR

MISO
SCK
NSS
PMIC_EMPTY
PMIC_RDY
PMIC_LOAD
MIC_ENB
SDA_B
SCL_B
D27|A5
D22|A0
D23|A1
D24|A2
D25|A3
D26|A4

MOSI
SCL_A
SDA_A
RX0
TX0
R7
4.7K
UPDI
RESET
DIO2
DIO1
SXRST
DIO0
LED2
YLED0603R
1k
R8
MIC_ENA
C18 0.1u
C19 0.1u
C20 0.1u
GND

XL1276 LoRa

U6
SX1276S9S+T-X1

MISO
MOSI
SCK
NSS
SXRST
DIO5
DIO2
DIO1
DIO0
C14 1n
C15 1u
ANT
3V3
GND

U.FL Connector

ANT
JP2
U.FL-R-SMT-1(80)
GND

MIC94069 LOAD SW

U8 MIC94069YML-TR
3V3
C16 1uF
VINA
ENA
VINB
ENB
VOUTA
VOUTB
MIC_ENA
MIC_ENB
C17 10u
C21 10u
VCC_SENS_B
VCC_SENS_A
GND

R9 10k
R10 10k
R11 10k
R12 10k
3V3
SCL_A
SDA_A
SCL_B
SDA_B

ENVIRONMENTAL SENSOR HDR

J3
HDR-M-2.54_2x2
VCC_SENS_A
GND

CO SENSOR HDR

J4
HDR-M-2.54_2x2
VCC_SENS_B
SDA_B
SCL_B
GND

UPDI AND SERIAL INTERFACES

H4 A2541WV-6P
RX0
UPDI
RESET
TX0
C27 10u
3V3
GND

MCU ANALOG INTERFACES

H5 A2541WV-6P
D22|A0
D23|A1
D24|A2
D25|A3
D26|A4
D27|A5

AEM10330 PMIC + PV HDR

U10
AEM10330-QFN
U9
KF301-5.0-2P
22u C22
L2 10u
VINT
VSTO
3V3
47u C25
U11
KF301-5.0-2P
10u C26
10u C24
CS_IN
ZMPP
SRC
BUFSRC
LIN
LOUT
R_MPP[2]
R_MPP[1]
VINT
R_MPP[0]
LOAD
BAL
STO
EN_STO_CH
STO
SLEEP
STO_OVDIS
STO_RDY
LOAD
CFG[2]
CFG[1]
CFG[0]
MPP[1]
STO_FT
MPP[0]
PRIO
HP
STO_CFG[3]
LOAD_CFG[1]
LOAD_CFG[0]
STO_OVCH
STO_RDY
STO_OVDIS
PMIC_EMPTY
PMIC_RDY
PMIC_LOAD
GND

TITLE: SDP EH Sensor Node	REV: 1.0
EasyEDA	Company: Ahmad Saad / Sheet: 1/1
	Date: 2026-02-24 Drawn By: Ahmad Saad